0.63158

7. **Ans: B**

For satellite in geostationary orbit, period = 24 hours = 86400 s

$$\omega = \frac{2\pi}{86400}$$

gravitational force provides for centripetal force

$$\frac{GMm}{r^2} = mr\omega^2$$

$$r = \left(\frac{GM}{\omega^2}\right)^{\frac{1}{3}} = \left(\frac{(6.67\times10^{-11})(5.97\times10^{24})}{\left(\frac{2\pi}{86400}\right)^2}\right)^{\frac{1}{3}} = 4.2227\times10^{7}\,m$$

$$\omega = \frac{v}{r}$$

$$v = \omega r = \left(\frac{2\pi}{86400}\right)(4.2227\times10^{7}) = 3071\,m\,s^{-1}$$

Alternatively, a shorter method would skip calculating the radius *r*.

$$mr\omega^2 = \frac{GMm}{r^2}$$

$$r^3\omega^2 = GM$$

$$\times\omega^3 \text{ to both sides}$$

$$(r^3\omega^3)(\omega^2) = GM\omega^3$$

$$v^3\omega^2 = GM\omega^3$$

$$v = \sqrt[3]{GM\omega} = \sqrt[3]{(6.67\times10^{-11})(5.97\times10^{24})\left(\frac{2\pi}{24\times3600}\right)} = 3071\,m\,s^{-1}$$

8. **Ans: B**